We've initialized the environment, but we haven't initialized the agents on the grid. We now have to decide how many agents and where to place them on the grid for each team.

Listing 9.14. Adding the agents

```
hid_layer = 25
in_size = 359
act_space = 21
layers = [(in_size,hid_layer),(hid_layer,act_space)]
params = gen_params(2,in_size*hid_layer+hid_layer*act_space)
1
map_size = 30
width = height = map_size
n1 = n2 = 16
2
gap = 1
3
epochs = 100
replay_size = 70
batch_size = 25

side1 = int(math.sqrt(n1)) * 2
pos1 = []
for x in range(width//2 - gap - side1, width//2 - gap - side1 +
side1, 2): 4
    for y in range((height - side1)//2, (height - side1)//2 +
side1, 2):
        pos1.append([x, y, 0])

side2 = int(math.sqrt(n2)) * 2
pos2 = []
for x in range(width//2 + gap, width//2 + gap + side2, 2):
5
    for y in range((height - side2)//2, (height - side2)//2 +
side2, 2):
        pos2.append([x, y, 0])

env.reset()
env.add_agents(team1, method="custom", pos=pos1)
6
env.add_agents(team2, method="custom", pos=pos2)
```

- ***1*** Generates two parameter vectors to parameterize two DQNs
- ***2*** Sets the number of agents for each team to 16
- ***3*** Sets the initial gap distance between each team's agents
- ***4*** Loops to position agents on team 1 on the left side of the grid